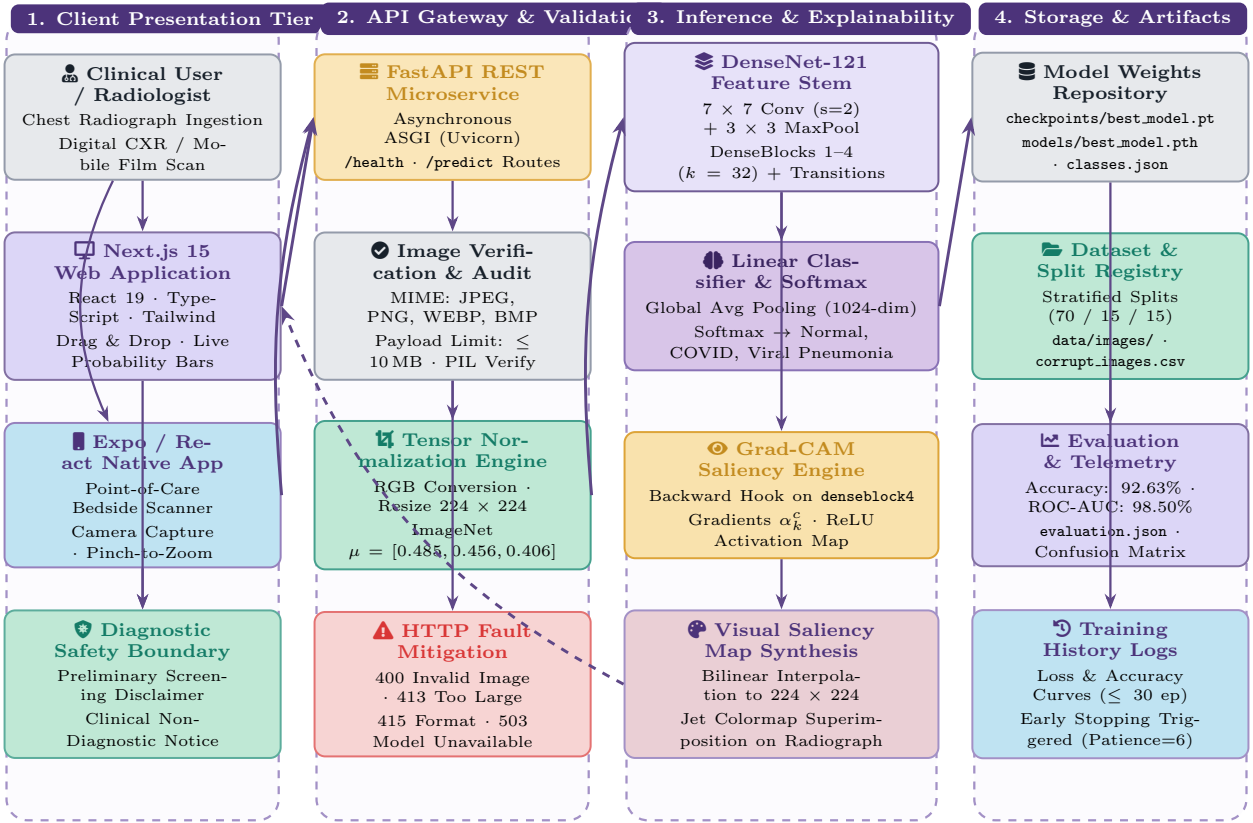

1. Client Presentation Tier
2. API Gateway & Validati
3. Inference & Explainability
4. Storage & Artifacts
Clinical User / Radiologist
Chest Radiograph Ingestion
Digital CXR / Mobile Film Scan
Next.js 15 Web Application
React 19 · TypeScript · Tailwind
Drag & Drop · Live Probability Bars
Expo / React Native App
Point-of-Care Bedside Scanner
Camera Capture · Pinch-to-Zoom
Diagnostic Safety Boundary
Preliminary Screening Disclaimer
Clinical Non-Diagnostic Notice
FastAPI REST Microservice
Asynchronous ASGI (Uvicorn)
/health · /predict Routes
Image Verification & Audit
MIME: JPEG, PNG, WEBP, BMP
Payload Limit: ≤ 10 MB · PIL Verify
Tensor Normalization Engine
RGB Conversion · Resize 224 × 224
ImageNet
μ = [0.485, 0.456, 0.406]
HTTP Fault Mitigation
400 Invalid Image · 413 Too Large
415 Format · 503 Model Unavailable
DenseNet-121 Feature Stem
7 × 7 Conv (s=2) + 3 × 3 MaxPool
DenseBlocks 1–4 (k = 32) + Transitions
Linear Classifier & Softmax
Global Avg Pooling (1024-dim)
Softmax → Normal, COVID, Viral Pneumonia
Grad-CAM Saliency Engine
Backward Hook on denseblock4
Gradients α_k^c · ReLU Activation Map
Visual Saliency Map Synthesis
Bilinear Interpolation to 224 × 224
Jet Colormap Superimposition on Radiograph
Model Weights Repository
checkpoints/best_model.pt
models/best_model.pth · classes.json
Dataset & Split Registry
Stratified Splits (70 / 15 / 15)
data/images/ · corrupt_images.csv
Evaluation & Telemetry
Accuracy: 92.63% · ROC-AUC: 98.50%
evaluation.json · Confusion Matrix
Training History Logs
Loss & Accuracy Curves (≤ 30 ep)
Early Stopping Triggered (Patience=6)